Sanaul Haque

Kevin_Polocoser_Proposal

Assignment 206

Document Details

Submission ID
trn:oid:::3117:636680033

Submission Date
Oct 9, 2026, 10:50 PM GMT+6

Download Date
Oct 9, 2026, 10:54 PM GMT+6

File Name
Kevin_Polocoser_Proposal.pdf

File Size
160.6 KB

7 Pages

2,234 Words

13,846 Characters

19% Overall Similarity

The combined total of all matches, including overlapping sources, for each database.

Match Groups

12 Not Cited or Quoted 12%
Matches with neither in-text citation nor quotation marks

3 Missing Quotations 1%
Matches that are still very similar to source material

5 Missing Citation 6%
Matches that have quotation marks, but no in-text citation

0 Cited and Quoted 0%
Matches with in-text citation present, but no quotation marks

Top Sources

16% Internet sources
16% Publications
11% Submitted works (Student Papers)

Match Groups

- **12** Not Cited or Quoted 12%
  Matches with neither in-text citation nor quotation marks
- **3** Missing Quotations 1%
  Matches that are still very similar to source material
- **5** Missing Citation 6%
  Matches that have quotation marks, but no in-text citation
- **0** Cited and Quoted 0%
  Matches with in-text citation present, but no quotation marks

Top Sources

16% Internet sources
16% Publications
11% Submitted works (Student Papers)

Top Sources

The sources with the highest number of matches within the submission. Overlapping sources will not be displayed.

1	Publication	Debashish Sur, Nathan Smith, Peter F. Connors, William H. Blades et al. "Investiga...	2%
2	Publication	Yijia Chen, Haoyu Yang, Rui Han, Hanze Wei, Kaiyuan Sun, Zhenyu Ji, Jinsong Guo, ...	2%
3	Publication	Bahar Atik, Anupa Kumari, Christos Englezos, Melih Ogeday Cicek et al. "Chirality ...	2%
4	Internet	kmaasrud.com	1%
5	Internet	publikationsserver.tu-braunschweig.de	1%
6	Internet	journal.uinjkt.ac.id	1%
7	Publication	Francesco Mazzeo, Luca Di Napoli, Massimiliana Carello, Martin Prokop, Martin P...	1%
8	Student papers	The University of Manchester on 2022-09-09	1%
9	Internet	arxiv.org	1%
10	Publication	Rong Pan, Tao Yang, Kangwon Seo. "Planning Constant-Stress Accelerated Life Te...	1%

11 Internet

www.mdpi.com <1%

12 Internet

www.aimsciences.org <1%

13 Internet

assets-eu.researchsquare.com <1%

14 Student papers

Delhi Technical Campus (The Sunshine Educational & Development Society) on 20... <1%

15 Publication

Francesco Ciucci. "Revisiting parameter identification in electrochemical impeda... <1%

16 Publication

Hamed Moayyed, Hussein Abdeltawab, Rishivandhiga Jayakumar, Lin Ma. "Accele... <1%

17 Internet

repository.up.ac.za <1%

18 Student papers

German University of Technology in Oman on 2023-07-16 <1%

Full Legal Name: Kevin Polocoser

Institution: Colegiul National Stefan cel Mare

Category: Junior

Class: 12th Grade

Country: Romania

Submission Date: September 20, 2026

Registered Email: kevin.polocoser04@gmail.com

Research Topic: Electrical Engineering: Electrochemical Impedance Spectroscopy and Optimal Experimental Design

Title: Model-Robust Frequency Selection for Sparse Electrochemical Impedance Spectroscopy Under Equivalent Circuit Uncertainty

Research Problem

EIS measures how the electrical response of an electrochemical system changes with frequency. The spectra are usually interpreted with equivalent circuit models, and the fitted parameters provide simplified representations of processes such as charge transfer, diffusion, or interfacial behavior. These quantities are reconstructed from the measured response rather than observed directly, so the problem is an inverse one. A fitted circuit should therefore reproduce the measured spectrum while yielding parameters that are meaningfully constrained by the available data [1], [2].

Full EIS measurements can span several orders of magnitude in frequency, and the low-frequency points take much longer to acquire than the high-frequency ones [2], [3]. This has led to sparse, optimally chosen frequency sets that try to keep the parameter information while cutting measurement time. Optimal experimental design has been used to find the frequencies that are most informative for parameter estimation, which could make battery, corrosion, and other impedance-based measurements considerably faster [3], [4].

A limitation of this approach is that these frequencies are usually optimized for one assumed equivalent circuit model [3], [4]. In practice, the appropriate circuit structure may itself be uncertain and different candidate models can reproduce similar spectra over a limited frequency range [2]. A frequency set that is very informative for the parameters of one model may omit frequency regions in which competing models are most distinguishable. With sufficiently sparse data, this could give apparently precise parameter estimates from an inadequate circuit representation [2]. This study will look at when such failures occur, and whether frequency sets designed to stay informative across several plausible models make the estimates more robust when the circuit itself is uncertain.

Existing Literature

Choosing which frequencies to measure in EIS has been studied as a way to improve parameter estimation while cutting experiment time. Ciucci et al. used model-based optimal experimental design to select impedance measurements that reduce estimated parameter uncertainty or shorten the experiment [3]. Ciucci later revisited parameter identification and optimal experimental design in EIS [4]. More recently, Py et al. showed that the usual practice of measuring a fixed number of logarithmically spaced frequencies is not necessarily optimal and proposed an active-learning method that picks informative frequencies during acquisition [5]. Similar ideas have since been applied to battery equivalent circuit identification, where E and D optimal designs have reduced the number of frequency points while preserving information relevant to parameter estimation [6], [7].

Choosing the equivalent circuit is a separate problem in EIS, since different circuit structures can produce very similar impedance spectra [2], [8]. Recent work has therefore looked at ways to distinguish candidate circuit structures from measured data [8]. Zhang et al. used Bayesian inference to test several corrosion equivalent circuits and showed that, as frequency information was removed, inference went from accurate and confident, to confident but inaccurate, and finally to unreliable [2]. Other recent approaches have used data-driven relaxation analysis to distinguish circuits with similar responses [8]. Together, these results suggest that reducing the

available frequency information affects not only parameter precision but also whether the data adequately support the assumed circuit structure.
These two problems are usually handled separately. Model-based optimal experimental design typically picks measurements by assuming a particular model and parameter region, whereas model selection is applied after the data have been collected [9]. Work on experimental design more broadly has shown that an optimal design can lose its advantages when the assumed model is wrong, and this has motivated robust designs that allow for several possible models or for model error [10]. The effect of equivalent circuit misspecification on optimized sparse EIS measurements, however, appears to be less well characterized. In particular, it is unclear when a frequency set optimized for one circuit starts to give misleadingly confident inference because another plausible circuit describes the system better, or whether a design that stays informative across several candidate models can reduce this failure.

Research Question

When do frequency sets optimized for a single equivalent circuit model become unreliable under model misspecification, and can multi-model frequency selection improve robustness?

Methodology

Study Design and Data Generation

This study will use a quantitative computational design based on three physically motivated corrosion equivalent circuit models. The circuit structures and parameter ranges will follow those examined by Zhang et al.: a charge-transfer-controlled uniform corrosion model, a passivating system that includes a Warburg element to represent diffusion, and a coated metal system with an additional coating and pore resistance branch [2]. These models represent different corrosion processes but can produce overlapping impedance responses under some conditions.
Dense reference spectra will be generated across the published frequency range of 10^{-3} to 10^{5} Hz, using parameter values within the reported physical ranges [2]. The published datasets and code will first be used to verify the implementation [2]. Because the generating model and its parameters are known for every synthetic spectrum, reconstruction and model selection errors can be measured directly rather than inferred from fitting residuals alone. Controlled complex measurement noise will then be added at several relative levels to test how measurement quality affects robustness to model misspecification.

Frequency Selection Strategies

Three frequency selection strategies will be compared under identical measurement budgets. The first will use conventional logarithmically spaced frequencies as a baseline. The second will use a single-model optimal design, where frequencies are selected to provide strong parameter information under one assumed equivalent circuit. To avoid confusing ordinary parameter variation with structural model error, this design will be optimized across several plausible parameter combinations within the assumed model rather than around one nominal parameter set [3], [4].
Parameter sensitivities will be normalized before constructing the information matrix so that differences in parameter units and numerical scales do not dominate the design. For each candidate frequency set, an E-optimal criterion will evaluate the weakest parameter direction within each model [6]. Pairwise T-optimal discrimination will then evaluate how well competing circuit predictions remain separated after the parameters of the rival model are adjusted to

minimize that separation [11]. The resulting parameter estimation and model discrimination efficiencies will be normalized to their corresponding single-objective optima. The multi-model strategy will then select the frequency set that maximizes the minimum normalized efficiency across all model-specific parameter estimation and pairwise discrimination objectives, avoiding a design that performs well for parameter estimation while becoming weak at distinguishing competing circuit structures [9], [10], [11].

All strategies will use the same candidate frequency range and the same number of measurements. Several measurement budgets, ranging from strongly sparse to moderately dense, will be tested so that the effect of frequency reduction can be compared fairly. The tested budgets will remain large enough for all candidate models to be fitted and compared using AICc.

Model Fitting and Evaluation

For each generated spectrum, only the frequencies selected by a given strategy will be treated as available measurements during reconstruction. Every candidate equivalent circuit will be fitted to the same sparse dataset using weighted complex nonlinear least squares with multiple starting points [4]. Parameter bounds, weighting, convergence criteria, and optimization settings will remain identical across strategies.

9 Candidate models will then be compared using the small-sample-corrected Akaike information criterion, or AICc, which penalizes additional parameters rather than selecting the circuit with the smallest residual alone [12]. Because the generating model is known, the proportion of cases in which the correct circuit is selected can be measured directly.

When the fitted and generating models have the same structure, relative parameter error will also be evaluated. When their structures differ, direct comparison of all fitted parameters will not be used because the circuit elements do not necessarily represent identical physical quantities. These cases will instead be evaluated through model-selection accuracy and predictive performance. Each sparse reconstruction will also be tested against the noise-free dense spectrum generated by the known reference model at frequencies that were not used for fitting. This provides an independent test of whether a model that reproduces the selected sparse measurements also predicts the wider frequency response.

Reliability Analysis

The main independent variables will be the frequency selection strategy, number of measured frequencies, measurement-noise level, and whether the model used to design the frequency set matches the model that generated the spectrum. The main outcomes will be correct model selection rate, parameter error where direct comparison is meaningful, and full-spectrum prediction error.

For representative cases near the transition between reliable and unreliable reconstruction, parameter uncertainty and model selection stability will also be examined using repeated noise realizations or parametric bootstrap resampling. Particular attention will be given to cases where sparse measurements produce low fitting error and apparently stable parameter estimates even though the wrong circuit is selected or prediction over the dense spectrum is poor. These cases represent the misleadingly confident inference that the study is intended to detect.

The same generated spectra will be evaluated under each sampling strategy, allowing paired comparisons under identical conditions. The final analysis will map how reconstruction reliability changes with measurement budget, noise, and model misspecification, allowing conventional, single-model optimal, and multi-model robust frequency selection to be compared directly.

Limitations

The study will consider only a finite set of equivalent circuit structures and therefore cannot represent every type of model mismatch that may occur in experimental EIS. Synthetic noise will provide a controlled approximation of measurement uncertainty but will not capture all effects found in physical measurements, such as drift, nonlinearity, or instrumentation artifacts. The selected corrosion models also represent specific electrochemical systems, so the resulting failure boundaries should not be assumed to transfer directly to every battery, corrosion, or bioimpedance model. The study is therefore intended to test model-robust frequency selection within a controlled set of plausible circuit uncertainties rather than define a universal EIS sampling strategy.

Research Topic

This research focuses on the reliability of sparse EIS measurements when the equivalent circuit model used to design the experiment may be wrong. Reducing the number of measured frequencies can make impedance measurements much faster, which is useful in applications such as battery diagnostics, corrosion monitoring, and other systems where repeated EIS measurements may be needed [3], [5], [6], [7]. However, frequency selection methods are often optimized under one assumed circuit model, even though the appropriate circuit structure may not be known in advance [2], [3], [8]. The proposed study therefore examines not only whether sparse measurements preserve parameter information, but also whether they remain reliable when the assumed model is misspecified. By comparing conventional, single-model optimal, and multi-model robust frequency selection strategies, the study aims to identify when model-based sparsification becomes unreliable and whether a more robust design can reduce these failures. The results could provide practical guidance for designing faster impedance measurements without sacrificing the ability to detect when the assumed circuit model is inadequate.

References

[1] S. Wang, J. Zhang, O. Gharbi, V. Vivier, M. Gao and M. E. Orazem, "Electrochemical impedance spectroscopy," *Nature Reviews Methods Primers,* vol. 1, Art. no. 41, 2021, doi: 10.1038/s43586-021-00039-w.

[2] R. Zhang, D. Sur, K. Li, J. Witt, R. Black, A. Whittingham, J. R. Scully and J. Hattrick-Simpers, "Bayesian assessment of commonly used equivalent circuit models for corrosion analysis in electrochemical impedance spectroscopy," *npj Materials Degradation,* vol. 8, Art. no. 120, 2024, doi: 10.1038/s41529-024-00537-8.

[3] F. Ciucci, T. Carraro, W. C. Chueh and W. Lai, "Reducing error and measurement time in impedance spectroscopy using model based optimal experimental design," *Electrochimica Acta,* vol. 56, no. 15, pp. 5416-5434, 2011, doi: 10.1016/j.electacta.2011.02.098.

[4] F. Ciucci, "Revisiting parameter identification in electrochemical impedance spectroscopy: Weighted least squares and optimal experimental design," *Electrochimica Acta,* vol. 87, pp. 532-545, 2013, doi: 10.1016/j.electacta.2012.09.073.

[5] B. Py, A. Maradesa and F. Ciucci, "Gaussian processes for the analysis of electrochemical impedance spectroscopy data: Prediction, filtering, and active learning," *Electrochimica Acta,* vol. 439, Art. no. 141688, 2023, doi: 10.1016/j.electacta.2022.141688.

[6] V. Sovljanski, M. Paolone, S. Tant and D. P. Sainflou, "Optimizing Experiments for Accurate Battery Circuit Parameters Estimation: Reduction and Adjustment of Frequency Set Used in Electrochemical Impedance Spectroscopy," in *2024 IEEE Energy Conversion Congress and Exposition (ECCE),* pp. 1935-1942, 2024, doi: 10.1109/ECCE55643.2024.10861495.

[7] I. Aguilar, E. Zulueta, U. Fernandez and J. Olarte, "Accelerated Electrochemical Impedance Spectroscopy of LFP Modules Using Gradient-Based Sensitivity and D-Optimal Selection," *Batteries,* vol. 12, no. 2, Art. no. 71, 2026, doi: 10.3390/batteries12020071.

[8] B. Patel, A. Sorrentino and T. Vidakovic-Koch, "Data-driven analysis of electrochemical impedance spectroscopy using the Loewner framework," *iScience,* vol. 28, no. 3, Art. no. 111987, 2025, doi: 10.1016/j.isci.2025.111987.

[9] Y. Liu, P. K. Maini and R. E. Baker, "Optimal experiment design for practical parameter identifiability and model discrimination," *Mathematical Biosciences,* vol. 399, Art. no. 109710, 2026, doi: 10.1016/j.mbs.2026.109710.

12 [10] R. E. Tsirpitzi, F. Miller and C.-F. Burman, "Robust optimal designs using a model misspecification term," *Metrika,* vol. 86, no. 7, pp. 781-804, 2023, doi: 10.1007/s00184-023-00893-6.

10 [11] A. C. Atkinson and V. V. Fedorov, "Optimal design: Experiments for discriminating between several models," *Biometrika,* vol. 62, no. 2, pp. 289-303, 1975, doi: 10.1093/biomet/62.2.289.

6 [12] C. M. Hurvich and C.-L. Tsai, "Regression and time series model selection in small samples," *Biometrika,* vol. 76, no. 2, pp. 297-307, 1989, doi: 10.1093/biomet/76.2.297.